



TrioDocs

Version: 0.4.0

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<https://triodocs.org>

# Remote Control

# Remote Control Overview

Trio can accept remote commands from *Nightscout* or from *LoopFollow*. There are a variety of options, but the final control of whether remote commands will be enacted rests with the Trio user. They can enable or disable remote control.

 **Nightscout version must be 15.0.2 or newer**

To properly display the OpenAPS pill with Trio 0.5.x (or newer), your *Nightscout* version must be 15.0.2 (or newer). If you do not see the expected treatments or pills in the *Nightscout* dashboard, follow the steps to [Configure for OpenAPS](#).

The most powerful arrangement, for Trio 0.5.x (or newer), is to configure the *LoopFollow* app to use the Trio Remote Control (TRC) setting.

The limited use of remote control with *Nightscout*, i.e., entry of Carb Correction and Temporary Targets when Careportal is authenticated, continues to be supported with Trio.

| <i>Nightscout</i> URL or App | Options                                                        |
|------------------------------|----------------------------------------------------------------|
| Careportal                   | Carb Correction<br>Temporary Target<br>Temporary Target Cancel |

Additional remote capabilities are offered for Trio using the *LoopFollow* app with these versions:

- Trio 0.5.x (or newer)
- *LoopFollow* version 2.4.0 (or newer)

| <i>LoopFollow</i> Remote Type | Options                                                            |
|-------------------------------|--------------------------------------------------------------------|
| <i>Nightscout</i>             | Set and Cancel Temp Target                                         |
| Trio Remote Control           | Meal (Carbs or Carbs & Bolus)<br>Bolus<br>Temp Target<br>Overrides |

### ? How does this differ from Trio 0.2.x?

Trio can use *Nightscout* Careportal to enter `Carb Correction`, and start and cancel `Temporary Target`.

- This was available in Trio 0.2.x and continues to be available in Trio 0.5.x (or newer).

Trio 0.2.x supported other remote options (using announcements via Careportal).

- Those options were replaced by the more secure Trio Remote Control for Trio 0.5.x (or newer)
- **Using announcements to provide remote control of the Trio phone is no longer supported**

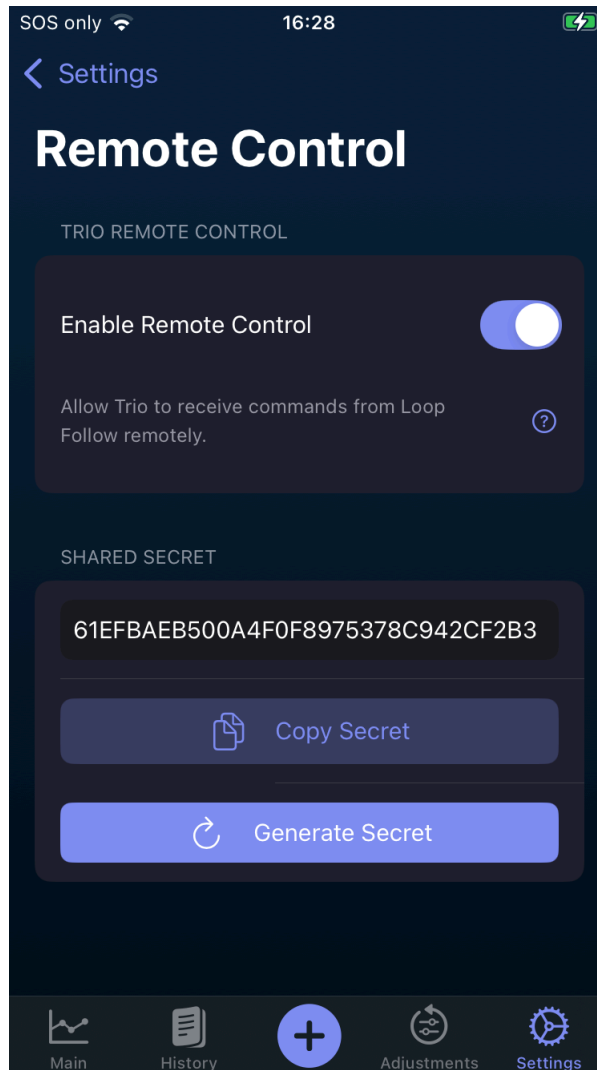
## Trio Remote Control

**Default:** *OFF*

Remote control must be enabled on the Trio phone or no remote information is accepted by the Trio phone.

You can search for this screen in Trio settings or go through the sequence: Trio, Settings, Features, Remote Control.

Once Remote Control is enabled, a Shared Secret is available. This is only used if you want to [Configure LoopFollow Trio Remote Control](#).



When Remote Control is enabled on the Trio app and the *LoopFollow* phone is properly configured, you can add carbs, send boluses, set or cancel overrides or temporary targets from the *LoopFollow* phone to the Trio phone via *Apple* push notifications.

The `SHARED SECRET` should be copied from the Trio phone and added to the `Shared Secret` row of the *LoopFollow* Remote Settings screen as part of the configuration for using *LoopFollow*.

#### Important

The ability for the Trio app to be remotely controlled will be **disabled** when `Enable Remote Control` is turned OFF, even if you have *LoopFollow* configured with the correct shared secret or your *Nightscout* URL has Careportal access. This is for the protection of the Trio user, so that they **always** are the primary controller of their insulin dosing app.

## *LoopFollow* Overview

Experienced *LoopFollow* users should still read this section - it touches on important configuration requirements for remote control.

The graphic below shows the *LoopFollow* features on the main screen of the app.

Glucose

Glucose Trend

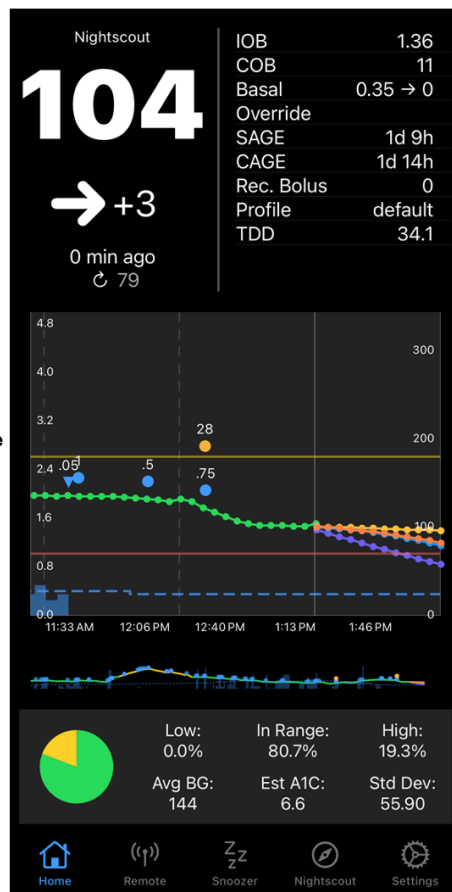
Last Loop Cycle  
Eventual Glucose

Scrollable/scalable  
graph display with  
BG, IOB, COB,  
Basal, Bolus, and  
the General  
Nightscout  
Careportal info

24-hour graph

24-hour results

Toolbar

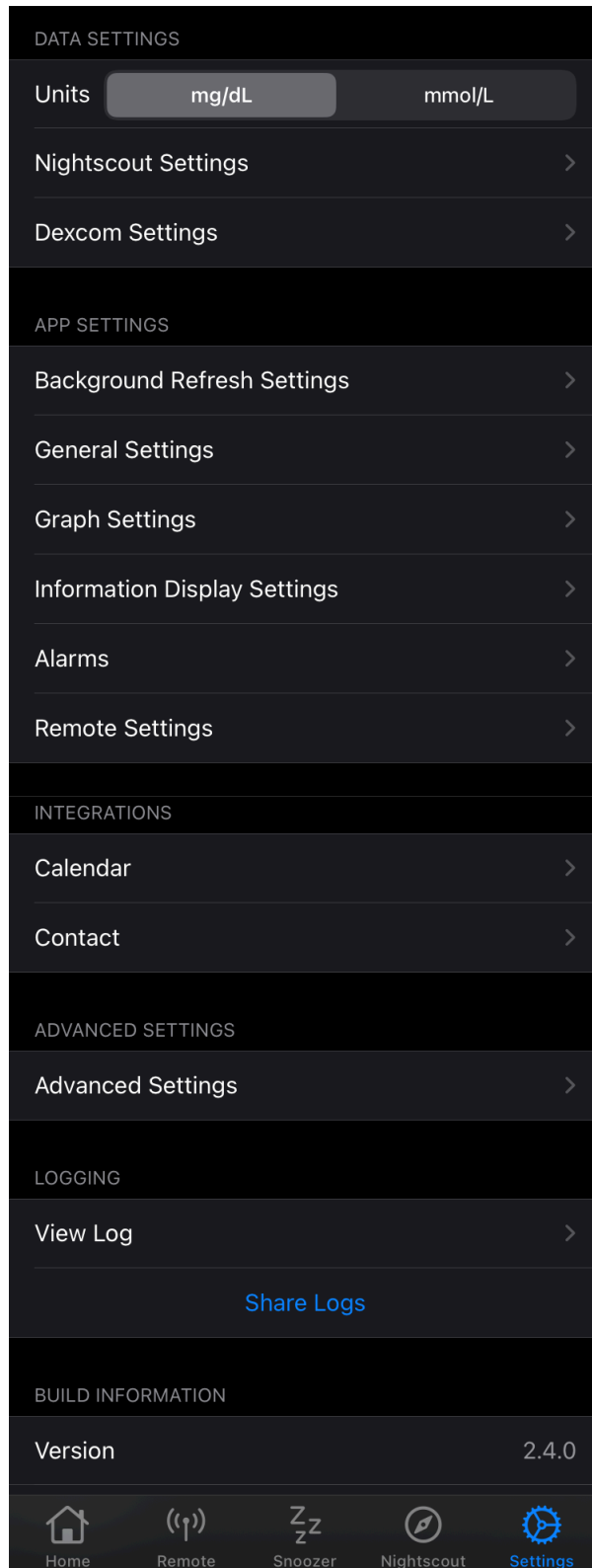


## Information Display Settings

|             |                                     |
|-------------|-------------------------------------|
| IOB         | <input checked="" type="checkbox"/> |
| COB         | <input checked="" type="checkbox"/> |
| Basal       | <input checked="" type="checkbox"/> |
| Override    | <input checked="" type="checkbox"/> |
| Battery     | <input type="checkbox"/>            |
| Pump        | <input type="checkbox"/>            |
| SAGE        | <input checked="" type="checkbox"/> |
| CAGE        | <input checked="" type="checkbox"/> |
| Rec. Bolus  | <input checked="" type="checkbox"/> |
| Min/Max     | <input type="checkbox"/>            |
| Carbs today | <input type="checkbox"/>            |
| Autosens    | <input checked="" type="checkbox"/> |
| Profile     | <input checked="" type="checkbox"/> |
| Target      | <input type="checkbox"/>            |
| ISF         | <input type="checkbox"/>            |
| CR          | <input type="checkbox"/>            |
| Updated     | <input type="checkbox"/>            |
| TDD         | <input checked="" type="checkbox"/> |
| IAGE        | <input type="checkbox"/>            |

## LoopFollow Settings

You tap on the settings icon at the bottom right of the toolbar to configure *LoopFollow*. The setting screen is shown in the graphic below.



## LoopFollow Data Source

You provide *LoopFollow* with information about the person you are following. At least one of these must be entered:

- [Nightscout URL](#)
- [Dexcom Share credentials](#)

### **Add *Nightscout***

## **Nightscout Access**

It is possible to have your Nightscout site readable by the world, in which case you do not need to add a token. If you choose to do that, just ignore references to entering the token below. The status will show up as `OK (Read)`.

The only exception is if you choose to [Use LoopFollow Nightscout Remote Control](#). In that case, you must have a token with careportal access.

For more information:

- [Admin Tools in Nightscout](#)

The Nightscout URL is required to enable display of the Information Table. It is recommended that you secure your Nightscout site so a token is required to view it. The type of token depends on the type of remote control desired. The table below indicates the minimum token access for each type of remote control available with Trio. When you enter your credentials, *LoopFollow* tries to reach the site and then provides the status.

| <i>LoopFollow</i> Remote Type | Minimum Token Access | <i>LoopFollow</i> Status |
|-------------------------------|----------------------|--------------------------|
| None                          | Read                 | OK (Read)                |
| <u>Nightscout</u>             | Read & Careportal    | OK (Read & Write)        |
| Trio Remote Control           | Read                 | OK (Read)                |

## **Do I need a token for Trio Remote Control**

The security for using *LoopFollow* Trio Remote Control comes from the Shared Secret and the APNS credentials. Your Nightscout site can be readable by the world if you so choose. In that case, no token is required and status will appear as `OK (Read)`.

The graphic below shows the display when you tap on the Nightscout Settings row.

To simplify setup, you can copy your Nightscout URL (including the token) from the [Admin Tools in Nightscout](#). When pasted into *LoopFollow* URL row, the app will automatically extract and fill in both the URL and token.

### Add Dexcom

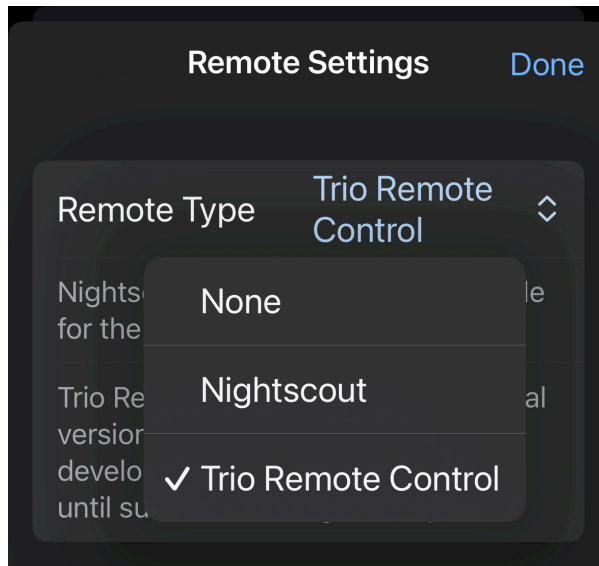
The graphic below shows the display when you tap on the *Dexcom* Settings row.

The *Dexcom* Share credentials are optional, but can be useful when the *Nightscout* URL is unavailable.

### LoopFollow Remote Setting Type

The Remote Settings row in the *LoopFollow* Settings screen is used to select the type of remote access you wish to use.





### The Trio Remote Control option is not available

The `Trio Remote Control` option is only available in *LoopFollow* if you have already entered a [Nightscout URL](#) with a default profile recognized as a Trio profile. Review [Troubleshooting](#) for possible reasons for not seeing the option.

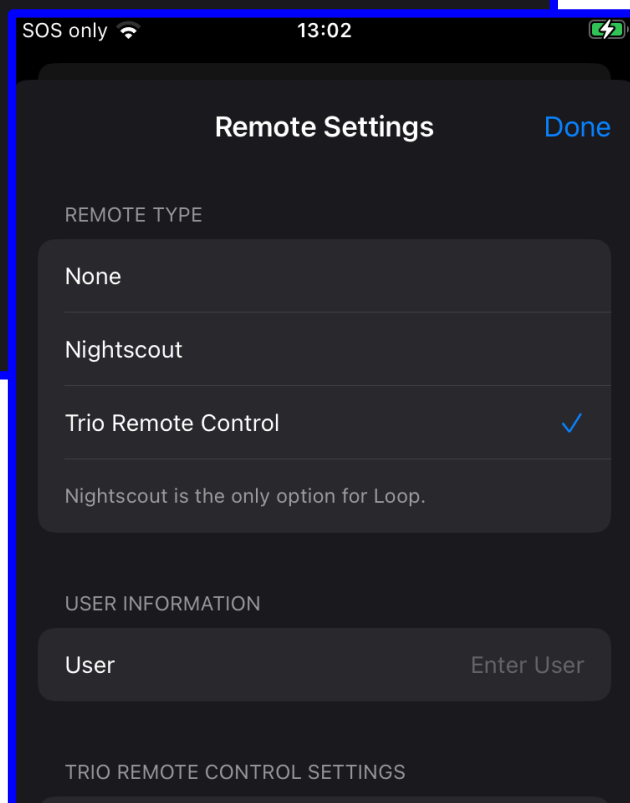
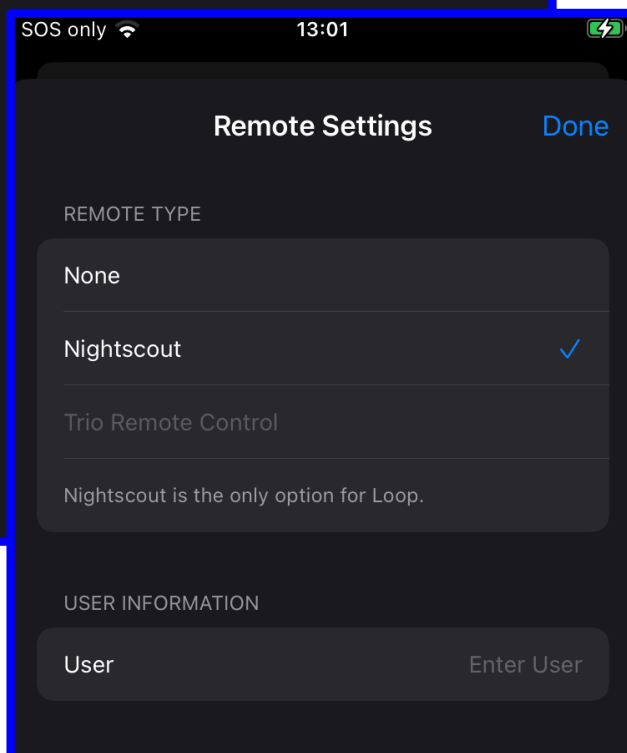
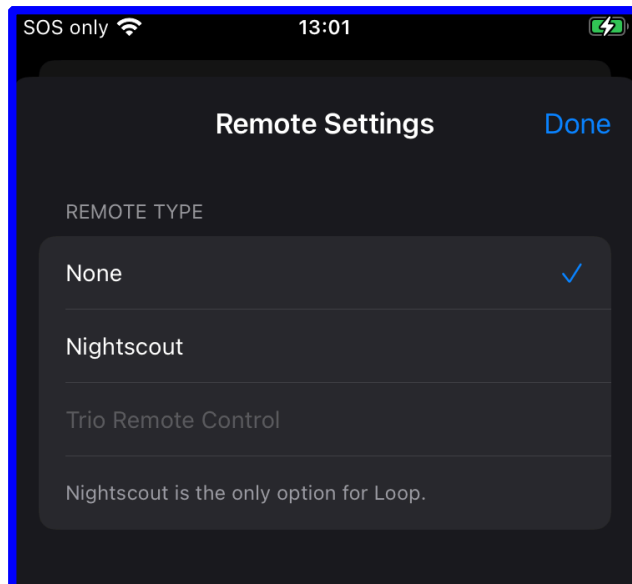
- [Nightscout](#) option
  - Remote control with *LoopFollow* is limited to starting and canceling Temp Targets
  - Available with Trio 0.2.x
- Trio Remote Control option
  - Remote control with *LoopFollow* includes adding remote carbs, enacting remote bolus, and starting and canceling Temp Targets and Overrides
  - Requires Trio 0.5.x (or newer) and *LoopFollow* 2.4.x (or newer)
  - Continue with [Configure LoopFollow Trio Remote Control](#) to finish the configuration process

## LoopFollow Remote Options

The *LoopFollow* app provides amazing display and alarm capabilities, whether remote control is enabled or not.

It can be used with the Trio or the [Loop](#) app with the `Nightscout` option for starting and stopping temp targets (Trio) or overrides (`Loop`).

The graphic below shows the top portion of the Remote Settings screen when `None`, `Nightscout` or `Trio Remote Control` is selected. The lower portion of the screen is found in the [Guardrails](#) section.



---

# Use *LoopFollow* *Nightscout* Remote Control

If you select *Nightscout* as the Remote Control Type for *LoopFollow*, this enables Temporary Targets to be set and disabled from *LoopFollow*.

This is the only remote option that works for Trio 0.2.x when using *LoopFollow*.

Remote

TEMPORARY TARGET

Target

120 mg/dL

Duration

30 minutes

Enact

Save as Preset

PRESETS

Eating Soon

Exercise

Test Preset

---

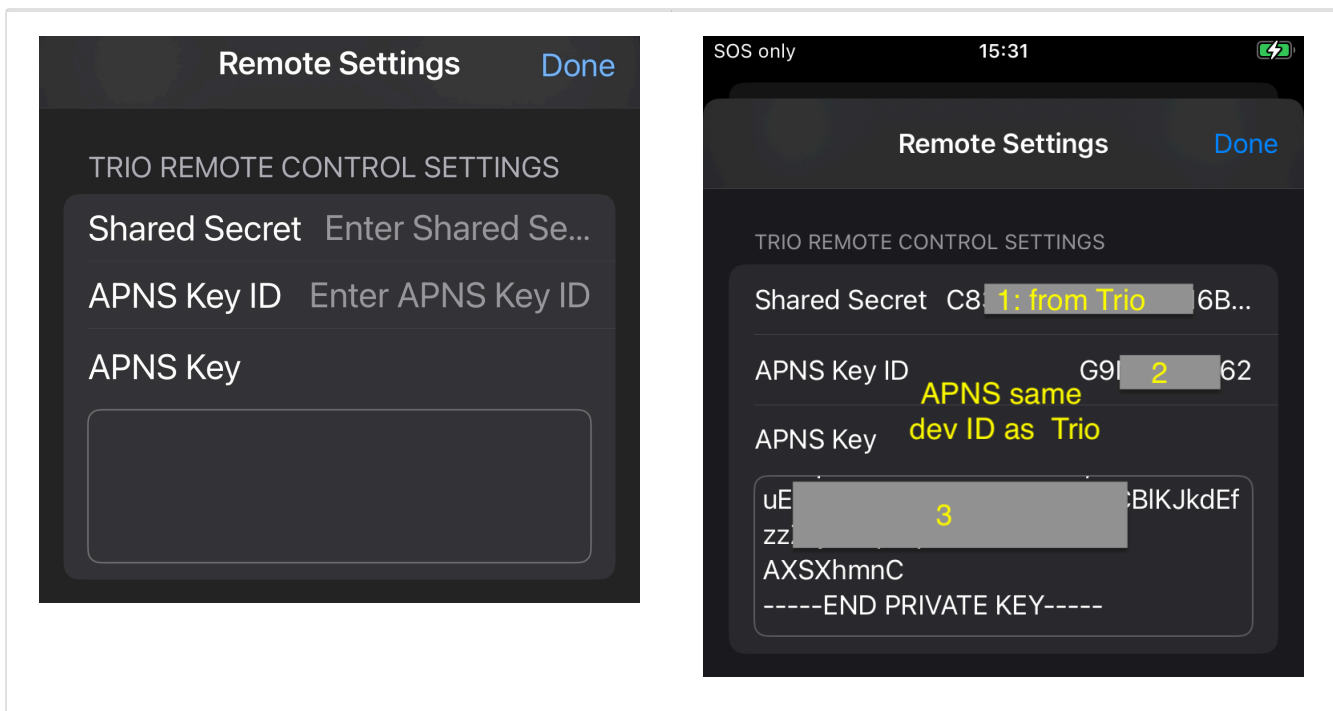
# Configure *LoopFollow* Trio Remote Control

This is supported for Trio 0.5.x (or newer) when using *LoopFollow* 2.4.0 (or newer).

When you select Trio Remote Control as the Remote Type in the *LoopFollow* app, you must fill in the (1) [Shared Secret](#), (2) [APNS Key ID](#) and (3) [APNS Key](#).

Default Remote Settings

Configured Remote Settings



## User

The person using the *LoopFollow* app should enter the name they want to show up as having entered this entry.

- At the current time, this is not used by *LoopFollow* for Trio
- It does show up as a notification on a *Loop* phone when *LoopFollow* is used with the *Loop* app
  - This feature might be added to *LoopFollow* for Trio at a later time

## Shared Secret

This is the unique shared secret that can be generated or entered into the Trio app in the Remote Control screen. The shared secret in Trio and *LoopFollow* must match to provide the ability to remotely send commands to this Trio app.

Please use a secure secret - the [automatically generated secret](#) is recommended.

## APNS Key ID

If you previously configured remote control with the *Loop* app, you already have an *Apple* Push Notification System (APNS) Key ID and Key. These were added to the config vars in your *Nightscout* site. See [Existing APNS](#). The value of the `LOOP_APNS_KEY_ID` goes here. Be sure to read the [Configure for OpenAPS](#) section about steps to make *Nightscout* and *LoopFollow* work with Trio.

If you have never created an APNS (or have lost the credentials), follow the directions in [New APNS](#) and copy the APNS Key ID into *LoopFollow* and save the value in your Secrets Reference file.

When creating the APNS, you must be logged in as a developer. The developer ID for the APNS must be the same as the one used for creating your Trio app or remote control will not work.

## APNS Key

If you previously configured remote control with the *Loop* app, you already have an *Apple* Push Notification System (APNS) Key ID and Key. These were added to the config vars in your *Nightscout* site. See [Existing APNS](#). The value of the `LOOP_APNS_KEY` goes here.

If you have never created an APNS (or have lost the credentials), follow the directions in [New APNS](#) and copy the APNS Key into *LoopFollow* and save the value in your Secrets Reference file.

## Guardrails

The maximum allowed entries for Bolus, Carbs, Protein, and Fat are configured in the guardrails section shown in the graphic below. This example is one in which the Shared Secret and APNS values have not yet been added.

| GUARDRAILS  |        |
|-------------|--------|
| Max Bolus   | 1.00 U |
| Max Carbs   | 30 g   |
| Max Protein | 30 g   |
| Max Fat     | 30 g   |

| MEAL SETTINGS         |                                     |
|-----------------------|-------------------------------------|
| Meal with Bolus       | <input checked="" type="checkbox"/> |
| Meal with Fat/Protein | <input checked="" type="checkbox"/> |

| DEBUG / INFO           |  |
|------------------------|--|
| Device Token:          |  |
| Production Env.: False |  |
| Team ID:               |  |
| Bundle ID:             |  |

## Meal Settings

The user can decide to enable or disable two features independently.

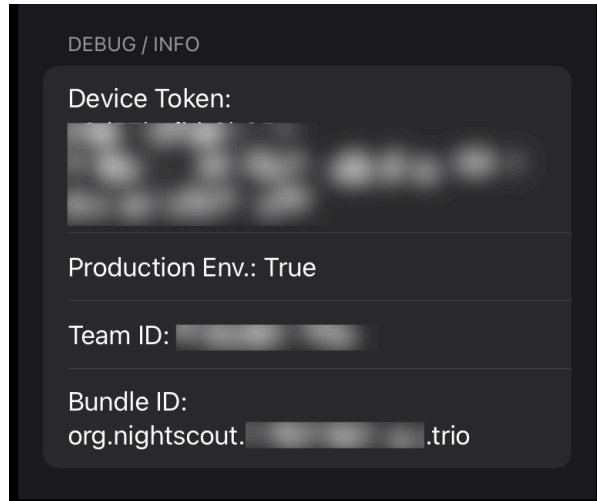
- Meal with Bolus
  - When enabled, a bolus command can be sent at the same time as the meal entry
- Meal with Fat/Protein
  - When enabled, the user is presented with a Protein and Fat row in addition to the Carbs and Bolus Amount rows

Refer to the graphic in the [Guardrails](#) section.

## Debug / Info

This section indicates if *Trio* has uploaded required information to *Nightscout*.

The graphic below shows a properly configured *LoopFollow* when the Trio app was built using the Browser Build method.



If you have empty rows in the Debug / Info screen, the most likely problem is the default profile is not coming from Trio. See [Update Profile](#). If you took those steps and still have missing rows, return to [Configure LoopFollow Trio Remote Control](#) and try again.

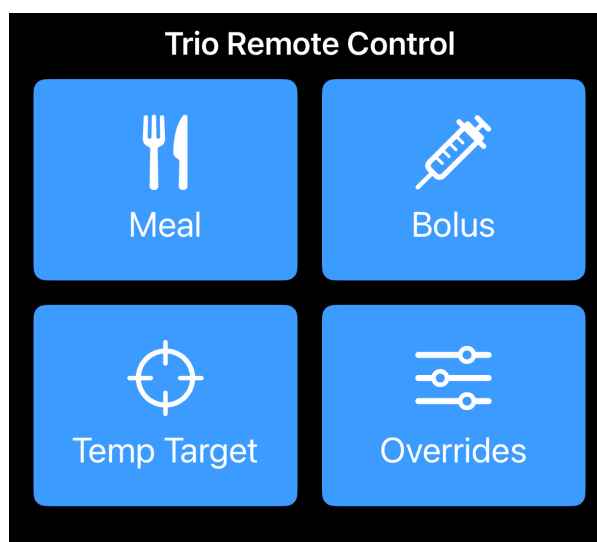
---

## Use *LoopFollow* Trio Remote Control

Once the *LoopFollow* phone is configured, and while the Trio phone is handy, test sending Remote Commands. It is good to also have a browser open with the *Nightscout* URL displayed.

Remember to give the system time to update.

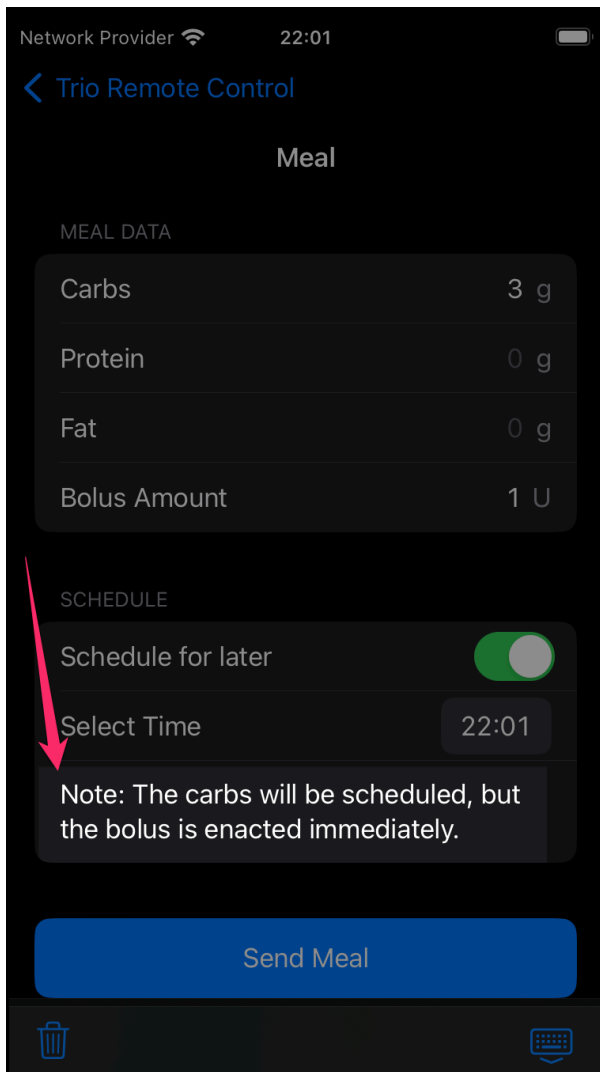
The sequence is *LoopFollow* to *Apple Push Notifications* to *Trio*, which uploads to *Nightscout* and then is displayed in the *LoopFollow* main screen.



## Remote Meal

**More info coming soon!**

When entering meals and choosing to schedule the meal, any bolus included in the meal is enacted immediately. Only the carb entry is entered according to the schedule.



Network Provider 22:01

< Trio Remote Control

Meal

MEAL DATA

|              |     |
|--------------|-----|
| Carbs        | 3 g |
| Protein      | 0 g |
| Fat          | 0 g |
| Bolus Amount | 1 U |

SCHEDULE

Schedule for later ☒

Select Time 22:01

Note: The carbs will be scheduled, but the bolus is enacted immediately.

Send Meal

## Remote Bolus

**More info coming soon!**

## Temp Target

**More info coming soon!**

## Overrides

**More info coming soon!**

---

## Apple Push Notifications System (APNS)

## Existing APNS

If you previously configured remote control with the *Loop* app, you already have an *Apple* Push Notification System (APNS) Key ID and Key. These were added to the config vars in your *Nightscout* site.

If you do not have existing APNS Keys, skip ahead to [New APNS](#).

When you configured APNS for the *Loop* app and saved information in your *Nightscout* config vars, they used the names in the table below. The same APNS Key ID and Key are what you need to add to the *LoopFollow* app in [Configure LoopFollow Trio Remote Control](#).

| Config Var       | Format of Config Var Value                                                                                                                                                                                                                                                     |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LOOP_APNS_KEY_ID | AAAAAAAAAA                                                                                                                                                                                                                                                                     |
| LOOP_APNS_KEY    | <pre> -----BEGIN PRIVATE KEY----- AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAA -----END PRIVATE KEY----- </pre> |

## New APNS

When using Trio, you do not need to add the config vars to *Nightscout* that are required for *Loop* remote control. If you already have them, it doesn't hurt anything, but you do not need to add them to use remote control with Trio.

When using Trio Remote Control with *LoopFollow*, the *Apple* Push Notification System is used directly. The Shared Secret provides additional security.

If you do not have APNS credentials, you need to create a key and grant it access to the Apple Push Notification Service (APNS).

Note - these directions are copied from *LoopDocs* so it suggests you name the key *Nightscout*. It is probably best to stick with that naming for APNS keys.

**Reminder**

This only works with the **paid** Apple Developer ID.

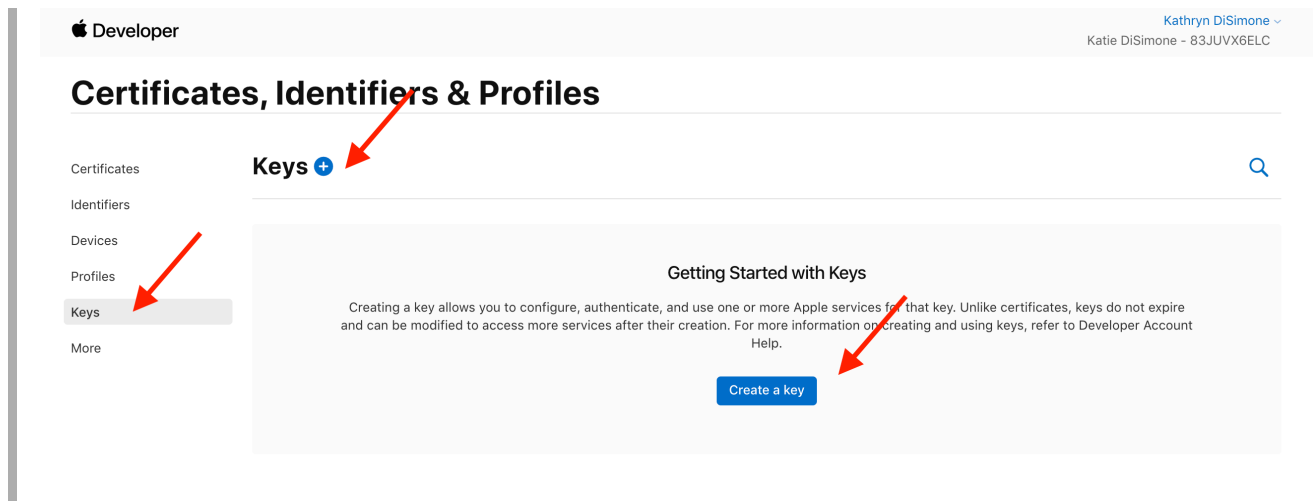


## ⚠️ Apple changed the APN system

Apple changed the way APN are created. Your old ones should still work, but if they don't, create new ones and update all the places where they are used.

When creating new APN keys, you have the option for "Sandbox", "Production" or "Sandbox & Production". Be sure to choose "Sandbox & Production".

1. To get started, go to the **Keys** section under Apple Developer's **Certificates, Identifiers & Profiles** and login with the *Apple ID* associated with your developer team that you used to build the *Trio* app.
2. If not already open in your browser (compare with the below screenshot),
  - Click on **Keys** (located in the left-hand column).
  - Either click on the blue **Create a new key** button **OR** the plus button (+) to add a new key.



3. In the form that appears, do the following:
  - Click the checkbox for enabling **Apple Push Notifications service (APNs)**
  - Enter a name for the key such as **Nightscout** (you can name it however you want, just make sure you know what the key is for by the name you choose).
  - Then click the **Configure** button to the right of the name
  - Choose **Sandbox & Production** and then **Save**
  - Tap on the **Continue** button, upper right

# Certificates, Identifiers & Profiles

[< All Keys](#)

## Register a New Key

Continue

Key Name

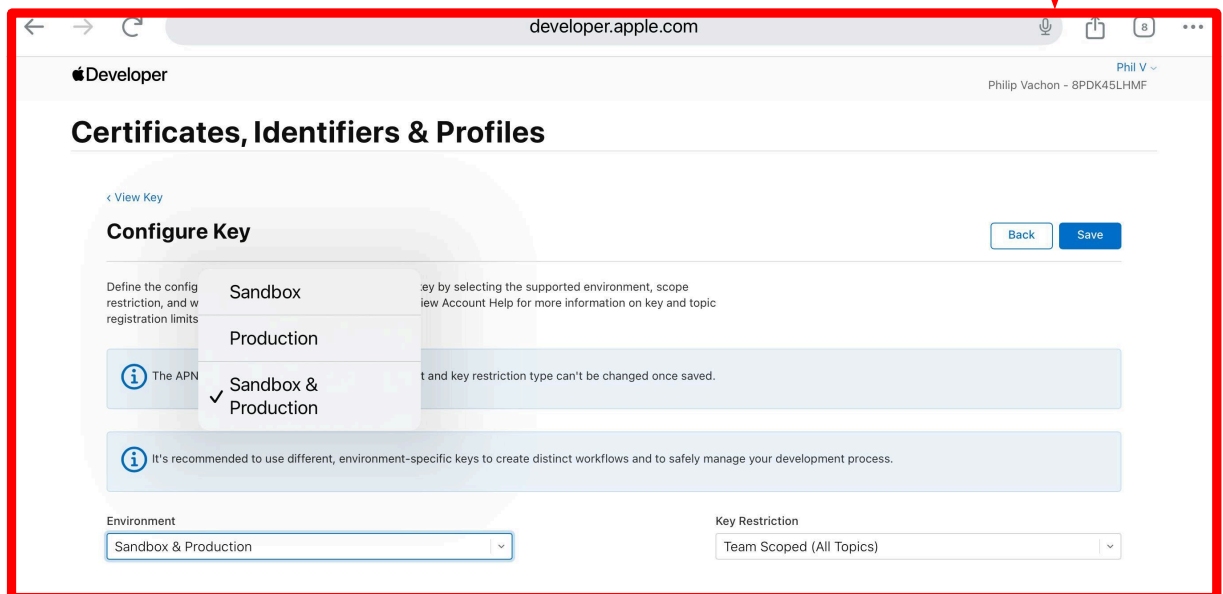
Nightscout

You cannot use special characters such as @, &, \*, !, ", -, .

Key Usage Description (optional)

| ENABLE                              | NAME                                                                                                                      | DESCRIPTION                                                                                                                                                                                                                                                    |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> |  Apple Push Notifications service (APNs) | Establish connectivity between your notification server and the Apple Push Notification service. One key is used for all of your apps. <a href="#">Learn more</a><br><span style="color: red;">ⓘ This service must have environment and type configured</span> |
| <input type="checkbox"/>            |  DeviceCheck                             | Access the DeviceCheck and AppAttest APIs to get data that your associated server can use in its business logic to protect your business while maintaining user privacy. <a href="#">Learn more</a>                                                            |

Configure



4. In the screen that follows, click the blue **Register** button.

Apple Developer

Kathryn DiSimone

Katie DiSimone - 83JUVX6ELC

## Certificates, Identifiers & Profiles

[< All Keys](#)

### Register a New Key

Back Register

Key Name  
Nightscout

| ENABLE                              | NAME                                    | SERVICE                                                                                                                                |
|-------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | Apple Push Notifications service (APNs) | Establish connectivity between your notification server and the Apple Push Notification service. One key is used for all of your apps. |

5. In the screen that follows, click the blue **Download** button.

This step will download a file with a name that starts with **AuthKey** and ends with **.p8**.

## Certificates, Identifiers & Profiles

< All Keys

### Download Your Key

Download

Done



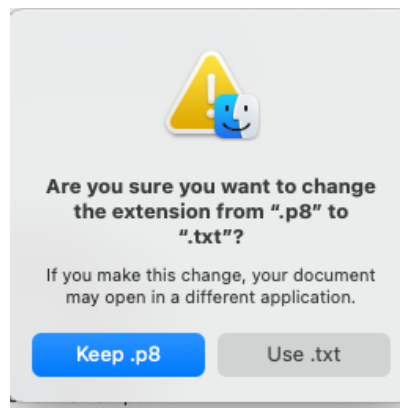
After downloading your key, it cannot be re-downloaded as the server copy is removed. If you are not prepared to download your key at this time, click Done and download it at a later time. Be sure to save a backup of your key in a secure place.

Name: Nightscout

Key ID: KPMAM8Y3G5

Services: Apple Push Notifications service (APNs)

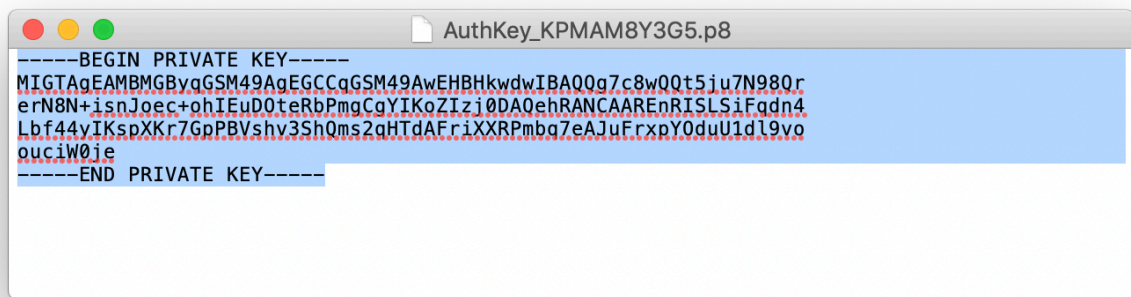
6. Find your `AuthKey` downloaded file in your downloads folder. It's a good idea to store this file where you can find it again if you need it. The next task is to rename the file so you can open it. Highlight the filename and choose rename, then add ".txt" after ".p8". In other words, modify `AuthKey_AAAAAAAAAA.p8` to `AuthKey_AAAAAAAAAA.p8.txt` and click on `Use .txt` when questioned.



7. Double-click to open the `AuthKey_AAAAAAAAAA.p8.txt` file. It will look similar to the screenshot below. You need to highlight **ALL OF THE CONTENTS** of that file and copy it and then paste it both into your Secrets Reference file and into the row for *LoopFollow* **APNS Key**. Yes, *alllllll* of the contents.

So, the easiest way is to:

- **Click inside that file**
- Highlight **all** the text, and then
- Copy **all** the text to the clipboard (Cf. screenshot below).
  - On a **Mac**, press `⌘ Cmd + A` to select all, then press `⌘ Cmd + C` to copy the selection.
  - On a **PC**, press `^ Ctrl + A` to select all, then press `^ Ctrl + C` to copy the selection.



8. The APNS Key ID is the 10-character name embedded in the filename: `AuthKey_XXXXXXXXXX.p8.txt`. You can also see it if you return to Apple Developer's [Certificates, Identifiers & Profiles](#) as highlighted in this graphic. You copy that APNS Key ID and then paste it both into your Secrets Reference file and into the row for *LoopFollow* [APNS Key ID](#)

## Certificates, Identifiers & Profiles

|              |        |          |  |  |  |  |  |
|--------------|--------|----------|--|--|--|--|--|
| Certificates | Keys + | Services |  |  |  |  |  |
| Identifiers  |        |          |  |  |  |  |  |
| Devices      |        |          |  |  |  |  |  |
| Profiles     |        |          |  |  |  |  |  |
| Keys         |        |          |  |  |  |  |  |
| Services     |        |          |  |  |  |  |  |

| KEY ID     | SERVICES | NAME | CREATED AT | UPDATED AT | APNS CONFIG              | APNS ENVIRONMENT     |
|------------|----------|------|------------|------------|--------------------------|----------------------|
| G9N9FR9H62 |          |      | 2022/11/15 | 2022/11/15 | Team Scoped (All topics) | Sandbox & Production |

## Troubleshooting

This section covers known troubleshooting issues:

- [Nightscout](#) not displaying Trio data: [Configure for OpenAPS](#)
- Was able to select Trio Remote Control in *LoopFollow* but it is no longer working: [Stop Nightscout access from the Loop app](#)
- Cannot select Trio Remote Control in *LoopFollow*: [Update Profile](#)

## Configure for OpenAPS

The [Nightscout](#) version must be 15.0.2 (or newer) to properly display the OpenAPS pill with Trio 0.5.x (or newer). Check your revision: [Nightscout](#) URL, Menu, scroll to bottom and examine the About section.

If you transitioned from the *Loop* app, you must make some modifications to [Nightscout](#) before you will be successful viewing your Trio data in your [Nightscout](#) site.

In [Nightscout](#), you need to modify these config vars:

| Config Var    | Loop | Trio    |
|---------------|------|---------|
| ENABLE        | loop | openaps |
| SHOW_PLUGINS  | loop | openaps |
| SHOW_FORECAST | loop | openaps |

Remember to restart the *Nightscout* server (restart dynos) after updating these variables.

## Stop *Nightscout* access from the *Loop* app

If you were previously running the *Loop* app:

- Remove *Nightscout* from *Loop* Services
- Add *Nightscout* credentials to Trio
  - You need the URL and the API\_SECRET.

In addition to this step, you may need to force the profile (from Trio) to upload to *Nightscout* and overwrite the one stored as the default profile in *Nightscout*.

## Update Profile

### Must on Trio 0.5.x (or newer)

If you are on Trio 0.2.x, you might see the option for Trio Remote Control in *LoopFollow* Remote Settings, but you can't use it. See [Use \*LoopFollow\* \*Nightscout\* Remote Control](#).

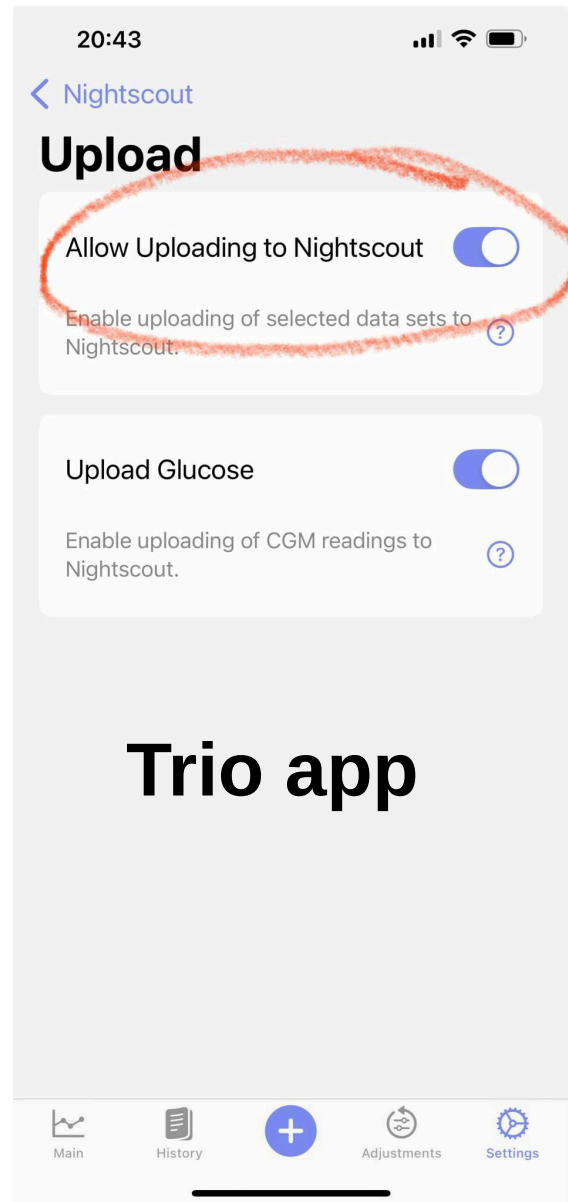
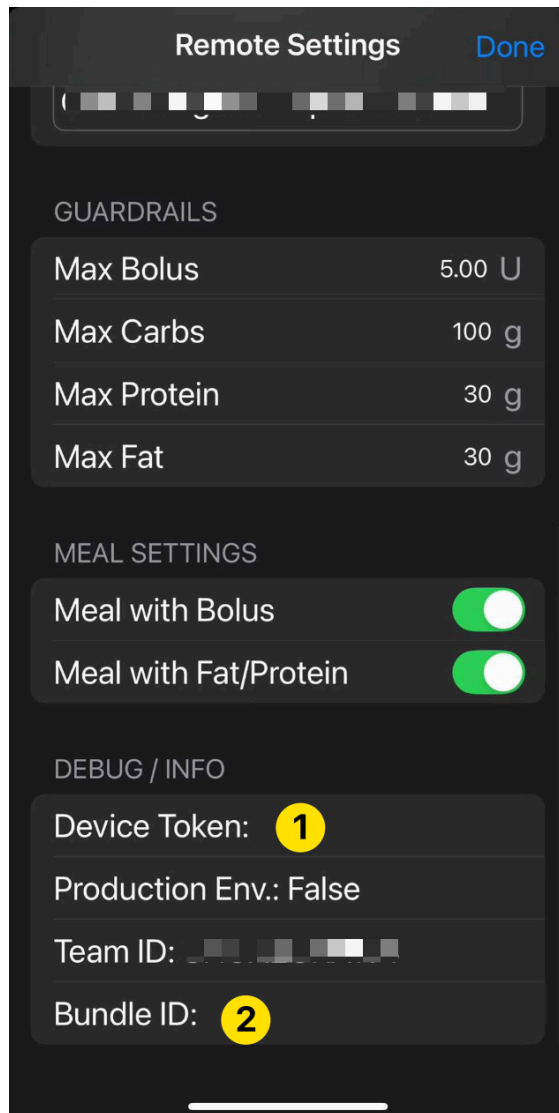
If you were previously running the *Loop* app, take the actions in the [previous section](#) and then force the profile to update.

To force a profile to update to *Nightscout*, go to the Trio app and toggle Allow Uploading to *Nightscout* off (disable) and then enable it again.

Once the user has toggled "Allow Uploading to *Nightscout*", *LoopFollow* needs to be refreshed (pull down glucose value to refresh) or re-started in order to fetch the correct information. *LoopFollow* will refresh eventually, but most users are impatient.

If the Debug Info in *LoopFollow* is missing a Device Token or a Bundle ID, as shown on the left side of the graphic, you need to make sure the *Loop* app is no longer uploading to *Nightscout* and force the profile to update.

# LoopFollow app



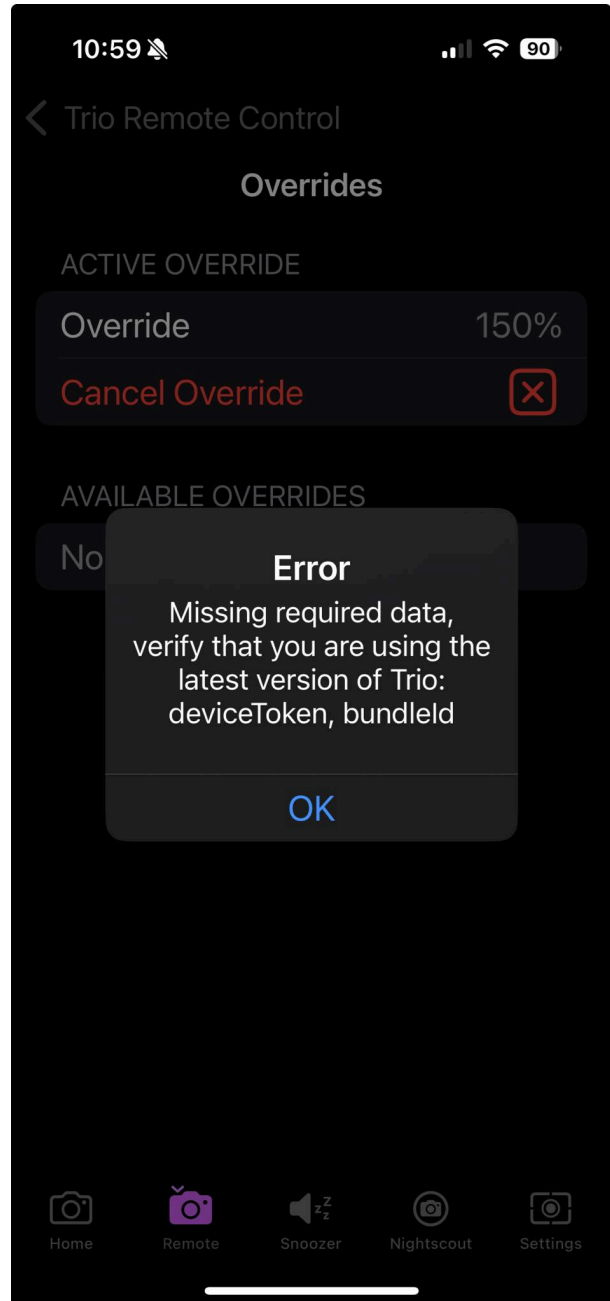
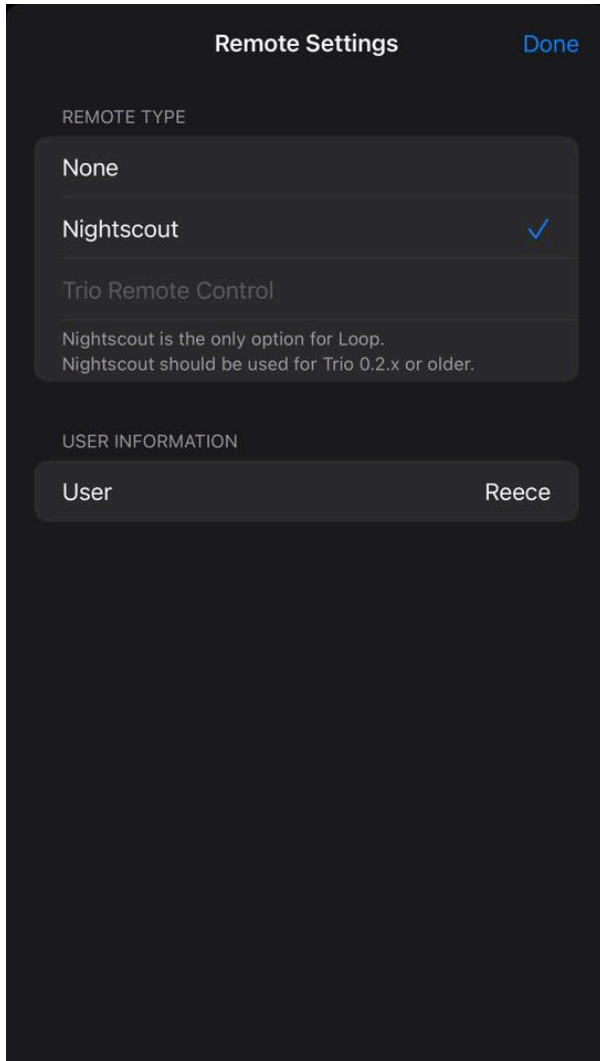
## Trio app

### Trio Remote Control Stops Working

Other signatures that you need to [force the update](#) are shown in the graphics below - for both these instances, Trio Remote Control (TRC) was working with *LoopFollow* and then stopped working:

TRC Option Not Allowed

TRC Error



## Build *LoopFollow*

This page is under construction. But here's the link:

[Install LoopFollow ↗](#)